Structural Patterns in News Headlines

Recovering form, framing, and rhetorical mode with interpretable rules

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Setup

The Question

Do news headlines follow stable syntactic templates

that can be modeled and evaluated rigorously?

Focus: linguistic form, not topic semantics.

Goal: interpretable structure + style prediction.

Why Headlines?

Headlines are extreme compression:

- Minimal length, maximal information
- Strong editorial constraints
- High public exposure and downstream impact

NLP relevance:

- Concise generation
- Summarization quality control
- Framing and agency analysis

Core Thesis

Headline writing is not free-form text production; it is a constrained system with reproducible structural regularities.

Hypothesis and Falsification

Hypothesis

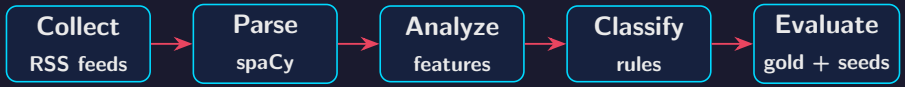
News headlines follow stable, reusable syntactic templates that are recoverable from parse-level features at rates above trivial baselines.

Falsification criteria:

- Macro F1 collapses toward majority/random behavior
- No dominant recurring templates in corpus-level analysis
- Large instability across random train/dev/test splits
- Rules cannot be linked to parse evidence

Method

Pipeline Overview



What flows through

- Tokens, POS tags, dependencies, entities
- Macro POS templates and openings
- Rule-priority structural label
- Multi-dimensional style profile

Parsing robustness

Model preference hierarchy plus normalization and a fallback parse pass for difficult casing and formatting conditions.

What “Gold” Means and Why We Need It

Gold-standard labels = human-verified ground truth

Why necessary:

- Prevent circular evaluation
- Measure true error, not self-agreement
- Enable valid precision/recall/F1
- Support fair model comparisons

Dataset design:

- 464 manually labeled headlines
- Train / Dev / Test = 276 / 90 / 98
- About 60 / 20 / 20 split, balanced for:
  - rule learning capacity (train),
  - tuning diagnostics (dev),
  - unbiased final estimate (test)

Without gold annotation, high scores can be meaningless.

Manual Tagging Examples

Example 1

- Headline: *Nine killed in second Turkish school shooting in two days*
- Gold label: `passive_clause`
- Gold rhetorical mode: `straight_report`
- Why: compressed passive event framing (`NUM + VBN`), omitted agent

Example 2

- Headline: *Iran Update Special Report, April 14, 2026 Institute for the Study of War*
- Gold label: `noun_phrase_fragment`
- Gold rhetorical mode: `analysis_explainer`
- Why: report-style nominal title with no finite main clause

Why These Metrics in This Study

Accuracy

- Fraction of headlines whose predicted label equals the gold label
- Useful for top-line editorial reliability

Precision$_c$ / Recall$_c$

- Precision$_c$: when we predict class c, how often are we right?
- Recall$_c$: of true class-c headlines, how many do we recover?

$F1_c$

- Harmonic mean of precision and recall for class c
- Punishes one-sided behavior

Macro F1

- Equal average over classes
- Critical under imbalance (‘simple_clause‘ dominates)
- Prevents "high accuracy, poor minority labels"

Structural and Style Label Space

Structure labels

- question_form
- passive_clause
- coordination
- noun_phrase_fragment
- simple_clause
- other

Style dimensions

- lead_frame
- agency_style
- density_band
- rhetorical_mode

$$\text{signature} = \text{structure} \mid \text{lead_frame} \mid \text{rhetorical_mode}$$

Evaluation Metrics

$$\text{Accuracy} = \frac{\#\text{correct}}{\#\text{total}}$$

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

$$F1_c = \frac{2PR}{P + R}, \quad \text{Macro F1} = \frac{1}{|C|} \sum_{c \in C} F1_c$$

Why Macro F1 matters

- Class imbalance is substantial
- Prevents dominant classes from hiding failures
- Better reflects minority-class behavior

In our context

- Accuracy = top-line headline match rate
- Macro F1 = class-balanced structural reliability

Worked Decision Paths (Classifier Logic)

Question-form example

- *Are we witnessing the death of expertise?*
- Path: contains "?" $\rightarrow$ interrogative rule fires first
- Output: `question_form`, `question_hook`

Passive compressed-event example

- *Nine killed in second Turkish school shooting in two days*
- Path: non-question $\rightarrow$ `NUM + VBN` passive cue
- Output: `passive_clause`, `passive_agent_omitted`

Worked Decision Paths (continued)

Nominal report-title example

- *Iran Update Special Report, April 14, 2026 Institute for the Study of War*
- Path: no question $\rightarrow$ no passive predicate $\rightarrow$ no finite clause
- Output: `noun_phrase_fragment`, `analysis_explainer`

Why this matters

These worked paths show that predictions are traceable to explicit rule triggers rather than opaque latent states.

Results

Headline Result

0.806

accuracy (test, $n = 98$)

+16.3 pp over majority

0.675

macro F1 (test)

0.130 majority $\rightarrow$ 0.675

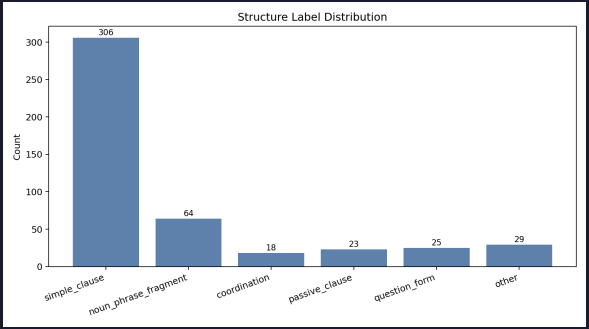
0.568

5-seed mean macro F1

$\pm$ 0.076 across splits

Headline form is recoverable, structurally constrained, and stylistically decomposable.

Corpus-Level Findings



Headline profile

- Avg length: 13.7 tokens
- Content-word ratio: 73.2%
- Headlines with verbs: 92.1%
- Active/non-passive: 94.6%
- Named entities present: 95.7%
- Actor/entity-first openings: 81.6%

Finding: headline form is compact, entity-heavy, and strongly action-oriented.

Model Performance Overview

Structure (held-out test)

- Accuracy: 0.806
- Macro F1: 0.675

Baselines (test)

- Majority: 0.643 acc, 0.130 macro F1
- Random: 0.459 acc, 0.159 macro F1

+16.3 pp acc over majority; +34.7 pp over random.

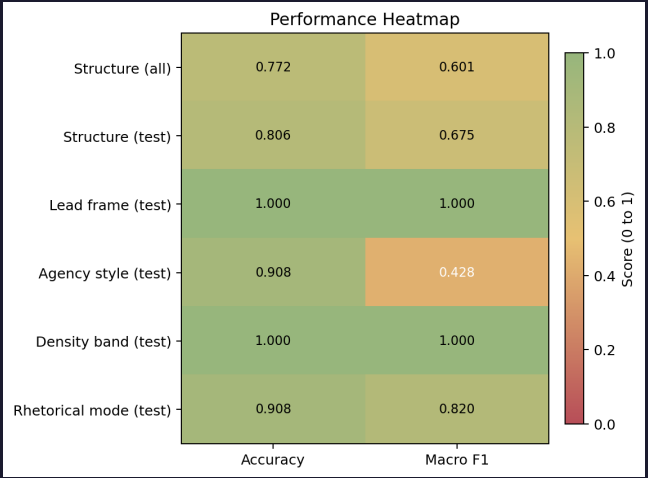
Style Performance (held-out test)

Dimension	Accuracy	Macro F1
lead_frame	1.000	1.000
agency_style	0.908	0.428
density_band	1.000	1.000
rhetorical_mode	0.908	0.820

Interpretation

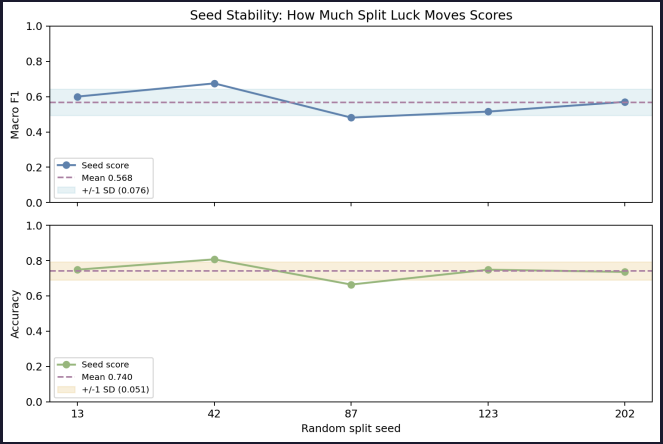
Density and framing are highly recoverable; rhetorical mode remains harder because it depends on broader discourse cues and boundary judgments.

Performance Heatmap



Compact view of performance across structure and style outputs.

Seeding and Split Luck



Why this analysis?

- One split can be lucky/unlucky
- Need expected performance, not one draw

5-seed summary

- Test accuracy: 0.740 ± 0.051
- Test macro F1: 0.568 ± 0.076
- Dev macro F1: 0.608 ± 0.073

Seeding and Split Luck (interpretation)

- Per-seed structure test macro F1 spans **0.481 to 0.675** (spread 0.194).
- Per-seed structure test accuracy spans **0.663 to 0.806** (spread 0.143).
- Macro F1 varies more than accuracy, so class-balance behavior is the more split-sensitive quantity.

Implication

Reporting only one split can overstate confidence. We therefore treat the seed mean as the expected score and the sample standard deviation as an uncertainty descriptor.

How We Quantify Split Variability

For metric values $m_1, \ldots, m_K$ over K random seeds:

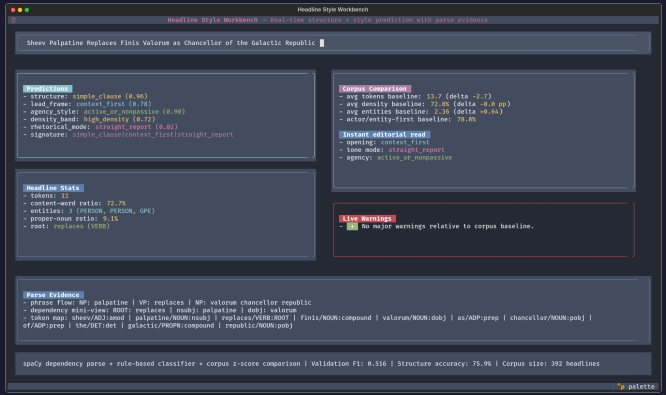
$$\bar{m} = \frac{1}{K} \sum_{i=1}^{K} m_i$$

$$s = \sqrt{\frac{1}{K-1} \sum_{i=1}^{K} (m_i - \bar{m})^2}$$

- $\bar{m}$ estimates expected performance under split randomness
- s measures split sensitivity (evaluation volatility)
- Per-seed trajectories reveal whether variance is smooth or driven by specific difficult splits

Practical Use

Interactive TUI Sandbox for Journalists



What it enables

- Real-time prediction + confidence
- Parse evidence per decision
- Instant headline A/B iteration
- Warnings for risky phrasing

A live sandbox for editorial A/B testing before publication.

Wrap-up

Discussion: What Worked and What Remains Hard

Strong signals

- Dominant templates are recoverable
- Multi-dimensional style profiling is informative
- Interpretability is operational, not cosmetic

Difficult regions

- Sparse "other" class behavior
- Coordination vs. simple-clause boundary cases
- Passive edge forms under punctuation/noise

Interpretation: gains are large versus baseline, but minority-structure distinctions remain the principal error source.

Limitations

- Rule-based systems are highly interpretable but less flexible than fully learned models
- Class imbalance can mask minority-class weaknesses
- Some labels remain intrinsically subjective at boundary cases
- Source and language scope limit broad external generalization
- No external lexical expansion in this phase (intentional scope constraint)

Despite these limits, structural signal remains strong and measurable.

Conclusion

Headlines are structured linguistic products

not free-form snippets

Structure test: 0.806 accuracy, 0.675 macro F1

with explicit split-luck quantification

Bottom Line

Interpretable NLP can capture headline form, evaluate it against human gold standards, and provide practical editorial tooling through a real-time sandbox interface.

“Headline form is measurable, predictable, and actionable.”

Victor Derani · Avi Herman · Kylie Lin

github.com/avih7531/headline-structure-analysis